

Using dynamic programming for making structured decisions about invasive species

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Invasive alien species (IAS) pose significant ecological, economic, and public health challenges. Robust methods are needed for guiding effective management. Overall, we hope to demonstrate the flexibility and utility of structured decision making in invasive species management.

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16 Introduction

17 Invasive species are a major global concern, with wide-ranging impacts on ecosystems,
18 economies, and public health ([Pyšek et al. 2020](#), [Roy et al. 2024](#)). Their management is
19 inherently challenging, as it requires repeated decisions under uncertainty, limited budgets,
20 and imperfect knowledge of ecological dynamics, which are often characterized by rapid
21 spread, strong population fluctuations, and broad spatial distributions. In practice, managers
22 must decide not only whether to intervene, but also when, where, and how intensively, while
23 balancing short-term operational costs against long-term socio-ecological benefits ([Hauser and](#)
24 [Possingham 2008](#)).

25 This makes invasive species management a fundamentally dynamic decision problem. Predict-
26 ing invasion dynamics or mapping risks is not sufficient unless these predictions are explicitly
27 linked to management objectives, feasible actions, and measurable consequences. Decision-
28 making is further complicated by the need to account for multiple, sometimes competing,
29 criteria, including ecological effectiveness, economic costs, operational feasibility, and potential
30 unintended consequences such as non-target impacts. In addition, management actions are em-
31 bedded in social contexts, where stakeholder perceptions, acceptability of control measures, and
32 underlying environmental values can strongly influence both implementation and outcomes
33 ([Simberloff et al. 2013](#), [Estévez et al. 2015](#), [Crowley et al. 2017](#))

34 Structured decision-making (SDM) and adaptive management (AM) have long been advocated
35 as frameworks to address the complexity of ecological management problems ([Williams et](#)
36 [al. 2002](#), [Thompson et al. 2021](#)). These approaches emphasize the explicit formulation of
37 objectives, the evaluation of alternative actions, and the incorporation of uncertainty into
38 decision processes. However, despite their conceptual appeal, they remain underused in
39 practice, in part because of the technical challenge of formally identifying, quantifying, and
40 optimizing trade-offs among competing objectives over time. Stochastic dynamic programming

(SDP), a core tool of decision theory, provides a powerful framework to address this challenge by treating management as a sequential decision problem under uncertainty. In this framework, optimal actions are derived as a function of the current state of the system (e.g. infestation size, spatial extent, or local density), while explicitly accounting for stochastic dynamics and the future consequences of present decisions. SDP naturally accommodates a wide range of management actions, including prevention, surveillance, containment, control, eradication, or inaction, and allows their outcomes to be evaluated in terms of long-term ecological, economic, or social objectives.

SDP has been successfully applied to a variety of invasive species management problems, including optimizing prevention versus control strategies for zebra mussels (Leung et al. 2002), designing control policies for invasive plants such as leafy spurge (Hyder et al. 2008), identifying state-dependent strategies for spatially structured invasions (Bogich et al. 2008), optimizing search effort for invasive insects (Baxter and Possingham 2011), and comparing alternative strategies across invasion stages (Hyytiäinen et al. 2013). More recent developments, including partially observable Markov decision processes (POMDPs), allow decision-making under imperfect detection, while connections with reinforcement learning highlight promising avenues for scaling these approaches to complex, data-rich systems (e.g. Haubrock et al. (2025)).

Despite these advances, the practical implementation of SDP and related approaches remains limited in applied animal ecology. This gap is partly due to the perceived complexity of these methods and the lack of accessible, step-by-step guidance for translating ecological problems into operational decision models.

In this paper, we aim to bridge this gap by providing a practical guide to the use of stochastic dynamic programming and its extensions for invasive species management. Using the coypu (*Myocastor coypus*) as a running example, we progressively illustrate how to formulate decision problems, implement models, and derive optimal management strategies across a

range of ecological contexts, from single populations to spatially structured metapopulations and interacting systems. We first introduce the core concepts of structured decision making and stochastic dynamic programming through a minimal, fully transparent example. We then develop a series of worked tutorials of increasing complexity, including spatial dynamics, imperfect detection using partially observable Markov decision processes (POMDPs), and model uncertainty addressed through adaptive management. Throughout, we provide reproducible implementations in R to facilitate uptake and adaptation by practitioners.

Explanation of the method

“A very brief historical background. Why is the method useful for investigating or analysing the topic in question? Provide some details on the method’s applications.”

We do not want to rewrite Lucile’s paper...

Structured decision making

Why and what (all)

Stochastic dynamic programming (Olivier)

Here we present a minimal stochastic dynamic programming (SDP) example. The aim is purely pedagogical: to show how states, actions, transition probabilities, rewards, and time horizon combine to produce an optimal decision rule.

We consider a stylized invasion management problem with three qualitative states describing the invasion level:

- Eradicated: no individuals detected (possible re-invasion risk)

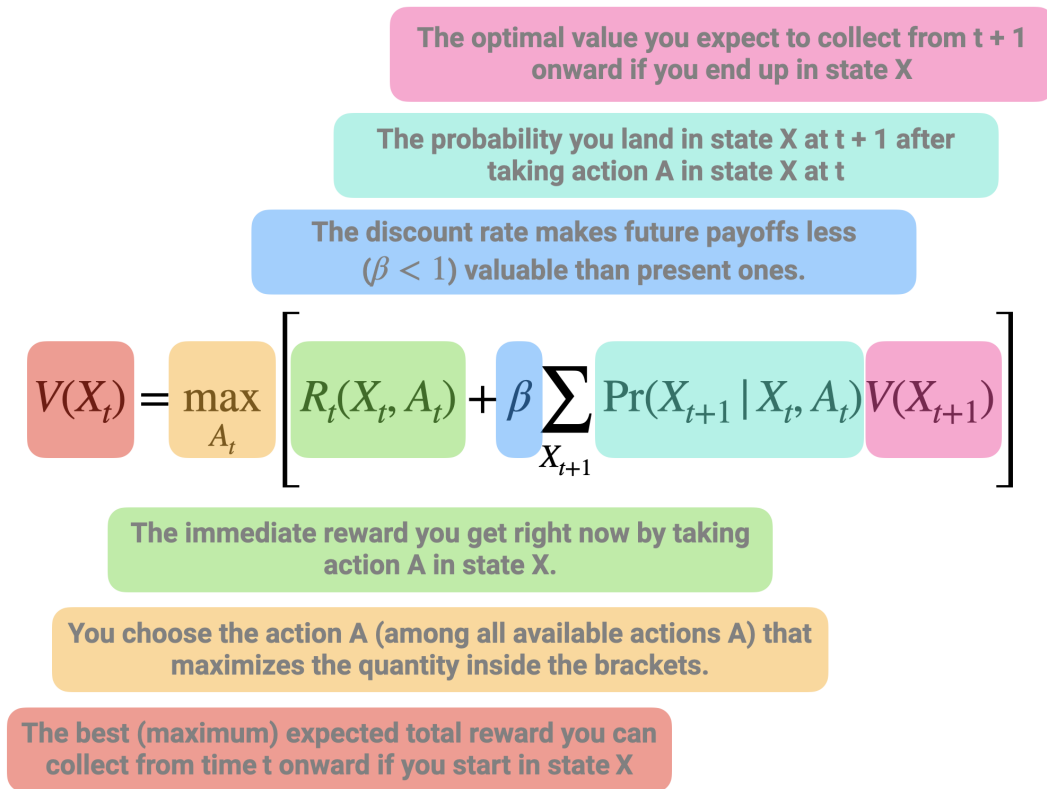


Figure 1: Schematic representation of the Bellman recursion in stochastic dynamic programming. The value of being in a given state at time t , $V(X_t)$, is defined as the maximum, over all possible actions, of two components: (i) the immediate reward obtained by taking action A_t in state X_t , and (ii) the expected discounted future value of subsequent states. The latter corresponds to the average (weighted by transition probabilities) of the optimal future values $V(X_{t+1})$ that can be reached from the current state-action pair, scaled by a discount factor $\beta < 1$ to reflect that future payoffs are less valuable than present ones. This recursive equation highlights the fact that the value of a decision is not only determined by its immediate payoff, but also by how it shapes future opportunities. The optimal action is therefore the one that maximizes the sum of current rewards and expected future benefits.

- Contained: invasion present but at low abundance
- Established: invasion widespread with high impacts

At each time step, the manager chooses between two actions:

- Monitor: surveillance only (low cost, little effect)
- Control: active management campaign (higher cost, improves state)

State dynamics are described by action-dependent transition matrices

$$P_a(s, s') = \Pr(S_{t+1} = s' \mid S_t = s, A_t = a).$$

Transition matrices describe how the invasion state evolves probabilistically from one time step to the next under each management action. Each row corresponds to the current state and each column to the next state, with entries giving transition probabilities. We build a transition matrix for each action.

Under monitoring, the transition is:

	Erاد.	Cont.	Estab.
Erاد.	0.95	0.05	0.00
Cont.	0.00	0.80	0.20
Estab.	0.00	0.00	1.00

If the system is eradicated, it remains so with probability 0.95 but may revert to a contained state with probability 0.05, reflecting a small risk of re-invasion. When the invasion is contained, it persists with probability 0.80 but may deteriorate to an established state with probability 0.20, indicating possible population growth without intervention. Once established, the system remains established with probability 1, meaning monitoring alone cannot reverse the invasion.

102 In contrast, control shifts the system toward better states:

		Erad.	Cont.	Estab.
$P_{\text{Control}} =$	Erad.	0.98	0.02	0.00
	Cont.	0.20	0.75	0.05
	Estab.	0.05	0.75	0.20

103 From an eradicated state, the system remains eradicated with probability 0.98 and only rarely
 104 becomes contained (0.02). When contained, control can eradicate the population (0.20), maintain
 105 it at low levels (0.75), or fail and allow establishment (0.05). Finally, when the invasion is
 106 established, control improves the situation in most cases, leading to a contained state with
 107 probability 0.75, occasionally achieving eradication (0.05), but sometimes failing to reduce
 108 impacts (remaining established with probability 0.20).

109 Together, these transition matrices encode both ecological uncertainty and the imperfect effec-
 110 tiveness of management actions.

111 Next we define rewards. Rewards are defined as the negative of total costs, so that maximizing
 112 expected reward is equivalent to minimizing management and damage costs.

$$R(s, a) = -(\text{damage}(s) + \text{cost}(a)).$$

113 Damage depends on the invasion state (higher when the invasion is established), while manage-
 114 ment cost depends on the chosen action (higher for active control than monitoring). The reward
 115 matrix is constructed by combining these two components for all state–action pairs, yielding
 116 the immediate payoff used in the dynamic programming recursion.

		Monitor	Control
$R =$	Erad.	-1	-6
	Cont.	-5	-10
	Estab.	-15	-20

117 (Rows correspond to the current invasion state and columns to the management action.)

118 For example, when the invasion is eradicated, monitoring yields a reward of -1, reflecting
119 only minor surveillance costs, whereas applying control in the same state yields -6 due to
120 higher intervention costs. When the population is contained, rewards are -5 for monitoring
121 and -10 for control, representing moderate damages combined with the respective management
122 costs. Finally, when the invasion is established, rewards drop to -15 under monitoring and
123 -20 under control, reflecting both high ecological impacts and, in the latter case, substantial
124 intervention effort. Overall, this reward structure captures the short-term trade-off between
125 ecological impacts and management effort that underlies the decision problem.

126 Solving the finite-horizon decision problem yields the optimal value function $V_t(s)$, which
127 represents the expected cumulative reward (i.e., negative total cost) obtained when starting
128 from state s at time t and following the optimal policy thereafter. Values are more negative
129 when the invasion is established, reflecting higher expected damages. Moreover, values increase
130 (become less negative) as the horizon shortens because fewer future costs can accumulate.

		$t = 1$	$t = 2$	$t = 3$	$t = 4$	$t = 5$	$t = 6$
$V_t(s) =$	$s = \text{Erad.}$	-7.60	-5.50	-3.69	-2.20	-1	0
	$s = \text{Cont.}$	-33.78	-27.43	-19.96	-12.0	-5	0
	$s = \text{Estab.}$	-49.26	-42.05	-34.47	-26.8	-15	0

131 The associated optimal policy $\pi_t(s)$ specifies the best action to take in each state s at each

time step t . Overall, the policy indicates that monitoring is typically optimal when the system is eradicated, whereas control is recommended when the invasion is established due to the high expected damages. In the contained state, the optimal decision varies with the time horizon, reflecting the trade-off between immediate intervention costs and the risk of future deterioration.

		$t = 1$	$t = 2$	$t = 3$	$t = 4$	$t = 5$
$\pi_t(s) =$	$s = \text{Erad.}$	Monitor	Monitor	Monitor	Monitor	Monitor
	$s = \text{Cont.}$	Control	Control	Monitor	Monitor	Monitor
	$s = \text{Estab.}$	Control	Control	Control	Control	Monitor

This toy example highlights key features of SDP:

- **State dependence:** optimal decisions depend on invasion level.
- **Trade-off:** control is costly but reduces future risk.
- **Time horizon effects:** as the horizon shortens, actions may become more short-term oriented.
- **Uncertainty:** decisions account for probabilistic transitions, not deterministic outcomes.

Things to consider for before using this method

“Outline the requirements for framing hypotheses, data and computing, experimental constraints and statistical power.”

Worked examples

“Papers should include at least two tutorials, with a step-by step demonstration of the method. Figures and mathematical formulations are particularly useful here. Where possible, tutorials

149 and examples should include those using the R statistical package or other freeware.”

150 Here explain that we use coyup IAS in Europe as a “fil rouge”.

151 **Single population (Olivier)**

152 The objective of this example is to determine an optimal regulation policy for coyup based on
153 their total abundance. We consider a stylized management problem in which a decision maker
154 chooses, at each time step, the intensity of a control effort $u \in [0, 1]$. Low effort corresponds
155 to minimal intervention, while high effort represents intensive removal. The management
156 objective is to balance three competing components: ecological damages that increase with
157 population size (e.g. crop losses, dike weakening, nuisance), management costs that increase
158 with effort, and an additional penalty if abundance exceeds a tolerable level.

159 In contrast to the previous example with discrete invasion states, the system is here described
160 by a continuous state variable, the population size N . Because dynamic programming requires
161 a finite state space, abundance is discretized onto a grid ranging from 0 to the carrying capacity
162 K . In the implementation, we set $K = 2000$ individuals and use a grid step of 20 individuals.
163 Control effort is also discretized, from 0 to 1 in increments of 0.05.

164 Population dynamics follow a density-dependent growth model with removals due to control,

$$N_{t+1} = N_t + rN_t \left(1 - \frac{N_t}{K}\right) - q(u)N_t,$$

165 where the first term describes logistic growth and $q(u)$ is the fraction of individuals removed as
166 a function of effort. We use a somehow saturating removal function,

$$q(u) = 1 - \exp(-\alpha u),$$

167 with $\alpha = 1.5$, which implies diminishing returns of control effort (Figure 2). For example, an

168 effort of $u = 0.5$ removes about 53% of the population, whereas maximum effort $u = 1$ removes
 169 about 78%, reflecting the fact that complete eradication is rarely achievable in a single step.

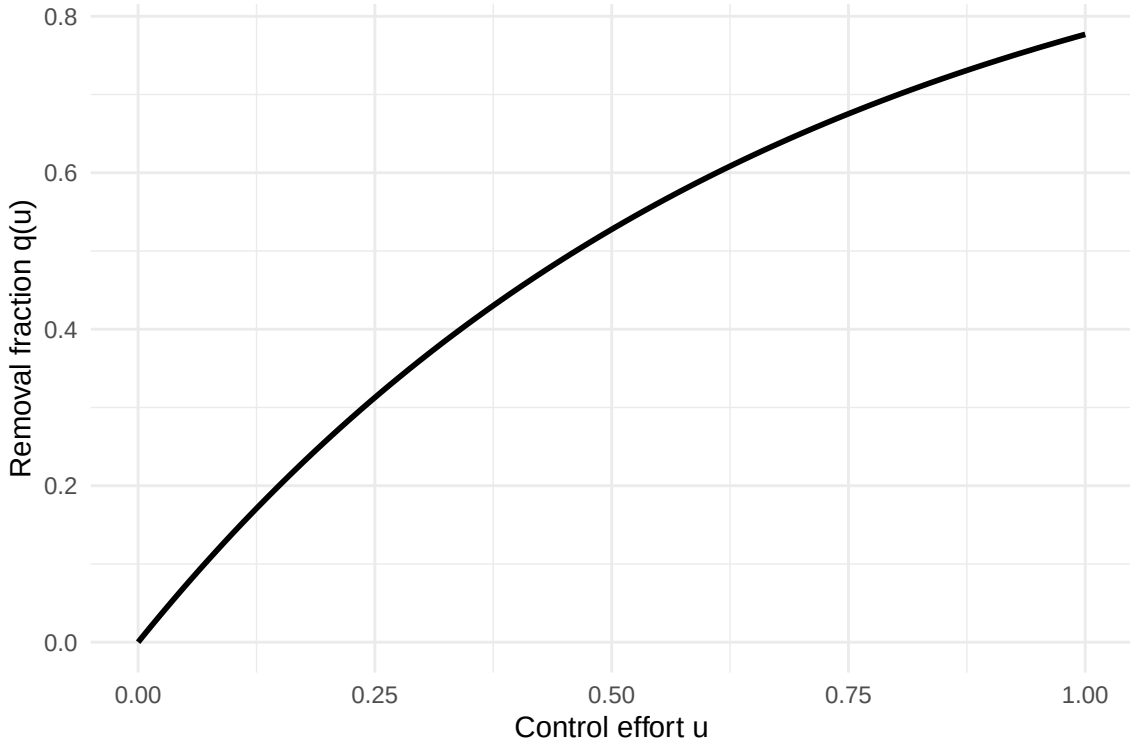


Figure 2: Removal fraction $q(u)$ as a function of control effort u .

170 Rewards are defined as the negative of total costs, so that maximizing expected reward is
 171 equivalent to minimizing long-term damages and management expenses (Figure 3):

$$R(N, u) = -(\text{damage}(N) + \text{cost}(u) + \text{penalty}(N)).$$

172 Damages increase non-linearly with abundance, here using a simple quadratic form
 173 $\text{damage}(N) = 0.1N + 0.001N^2$, reflecting the common ecological pattern whereby impacts
 174 accelerate at high density. Management costs increase with effort and are assumed convex
 175 ($\text{cost}(u) = 50u + 200u^2$), capturing increasing marginal costs of intensive interventions. Finally,
 176 a penalty is applied when abundance exceeds a tolerable threshold $N_{\text{tol}} = 400$, representing
 177 situations where ecological or social impacts become unacceptable. The penalty increases
 178 linearly beyond this threshold according to

$$\text{penalty}(N) = \begin{cases} 0, & N \leq N_{\text{tol}}, \\ 2000 \frac{N - N_{\text{tol}}}{N_{\text{tol}}}, & N > N_{\text{tol}}. \end{cases}$$

179 This formulation implies that the penalty is zero below the threshold and then rises proportion-
 180 ally to the exceedance. For example, an abundance of $N = 800$ yields a penalty of 2000, thereby
 181 strongly incentivizing control once populations grow beyond acceptable levels.

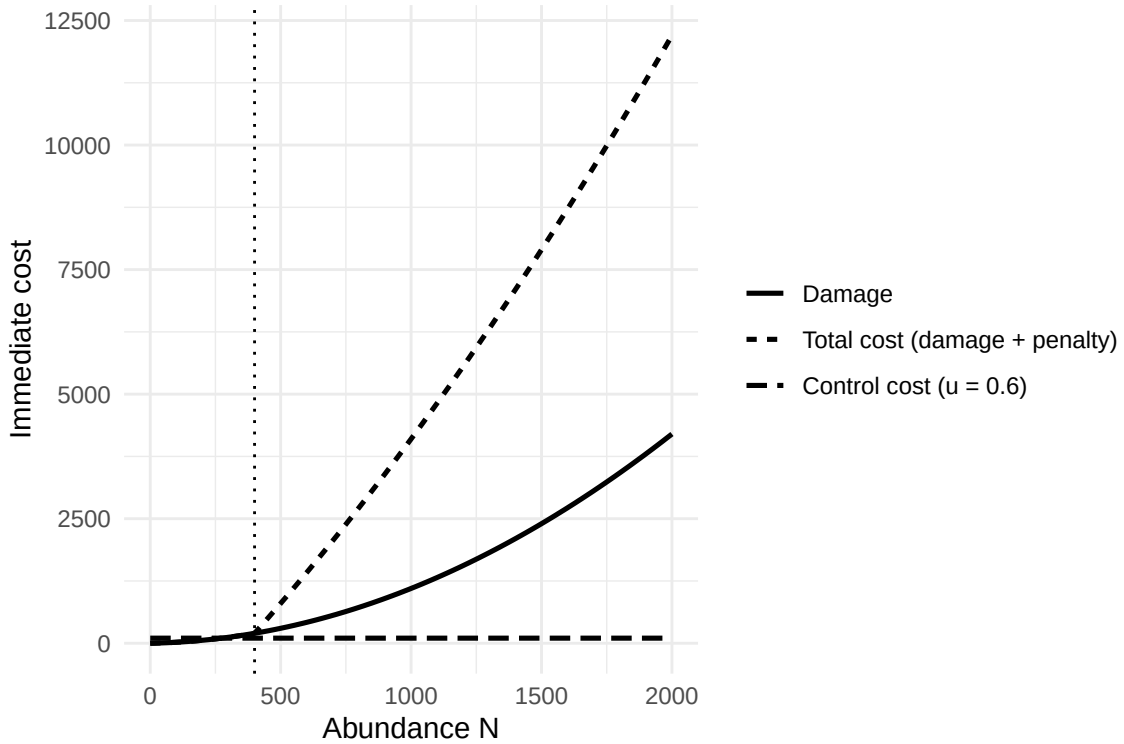


Figure 3: Immediate cost components as functions of abundance. Damage increases nonlinearly with population size, leading to rapidly rising total costs (damage + penalty). The dotted line marks the tolerable threshold ($N_{\text{tol}} = 400$) above which an additional penalty is applied in the decision model, although it is not shown here for clarity. Control cost is illustrated for an example effort level ($u = 0.6$).

182 The decision problem is solved using infinite-horizon discounted value iteration (with discount
 183 factor $\gamma = 0.99$). The resulting value function represents the expected long-term net benefit of
 184 starting from each abundance level when optimal decisions are applied thereafter. The optimal
 185 policy $u(N)$ typically exhibits a threshold-like pattern: low effort at low abundance where
 186 damages are limited, progressively stronger intervention as abundance increases, and intensive
 187 control once the population approaches or exceeds the tolerable level. This behavior reflects a

general ecological intuition — management intensity should scale with both population size and risk — and illustrates how stochastic dynamic programming formalizes this principle within a transparent decision framework.

The decision problem is solved using infinite-horizon discounted value iteration (discount factor $\gamma = 0.99$). This yields (i) an optimal stationary policy, mapping each abundance class to a recommended control effort, and (ii) a value function (not shown) that summarizes the long-term expected net benefit when the policy is followed.

Because damages and the penalty increase sharply with abundance, the optimal rule is “threshold-like” (top panel in Figure 4). Effort is low at small N , then increases rapidly as N approaches the tolerable threshold $N_{\text{tol}} = 400$, and remains high for large N . Ecologically, this tells us that when the population is small, impacts are limited and additional control has diminishing returns, whereas once abundance enters an unacceptable zone, strong intervention is warranted to prevent sustained high impacts.

The bottom panel of Figure 4 illustrates the dynamical consequences of applying this rule to a trajectory starting from $N_0 = 800$. Control effort $u(t)$ adjusts immediately to the current abundance class, and abundance $N(t)$ responds through the logistic dynamics with removals. In this deterministic grid-based version, the system typically moves quickly toward a lower-abundance regime and then stabilizes around a band where the marginal benefit of additional control balances its marginal cost.

Metapopulation (Olivier)

We extend the single-population coy-pu example to a spatially structured invasion in which several sites are connected through dispersal. The aim is to explore how management effort should vary through time when local demographic processes and spatial spread interact. We consider a network of several sites that can be interpreted as cities or management units. At

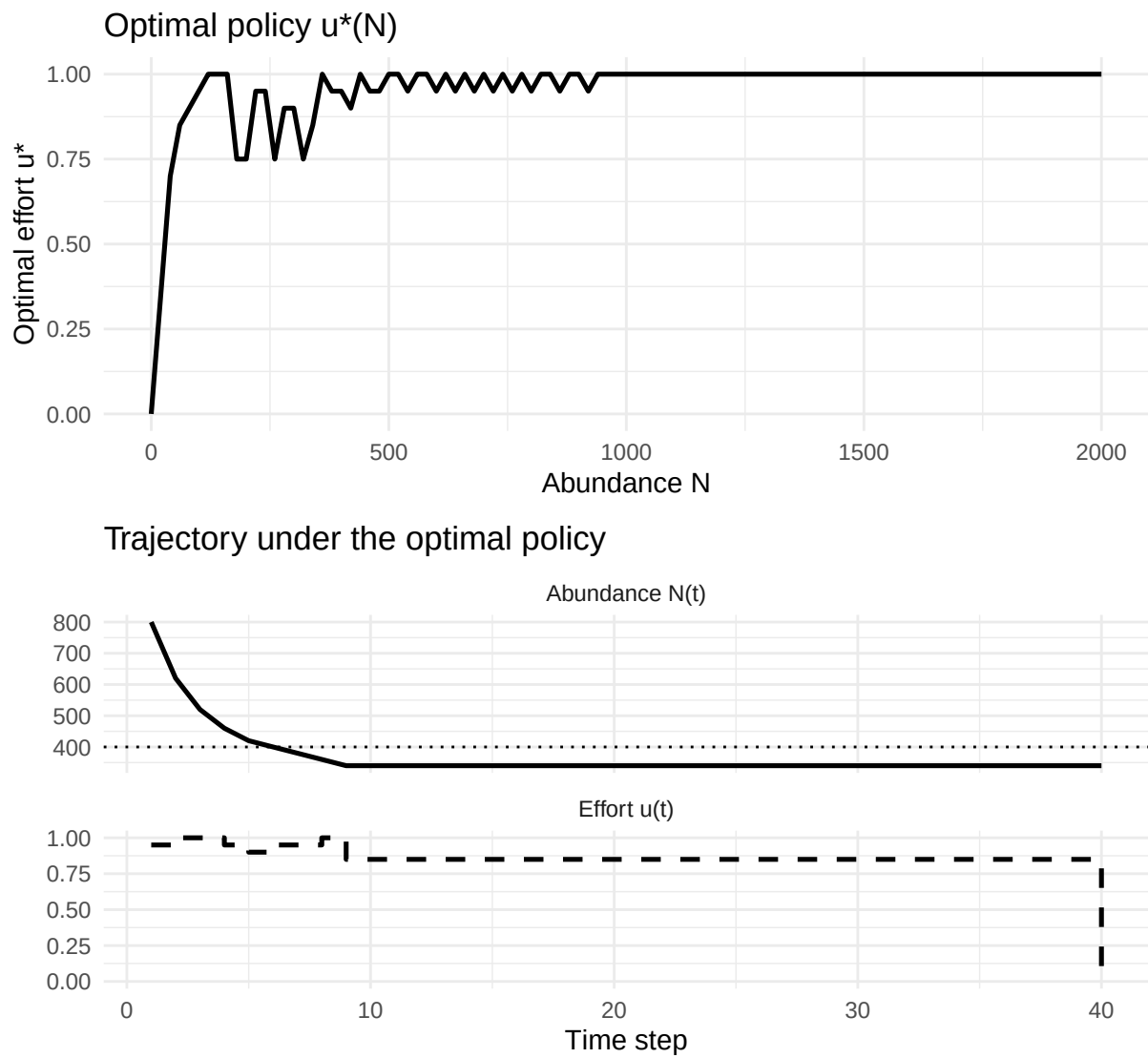


Figure 4: Optimal policy $u^*(N)$ (top) and an example trajectory under the optimal policy (bottom, faceted into abundance and effort), starting from $N_0 = 800$.

each monthly time step, a decision maker chooses a global management action corresponding to a harvest intensity applied simultaneously across all sites. Local populations then grow according to density dependence, are partially removed by management, and may subsequently undergo stochastic extinction or colonization depending on local abundance and the occupancy of neighboring sites.

At time t , each site i is either occupied or empty, represented by the binary variable $X_i(t) \in \{0, 1\}$. When a site is empty its abundance is zero, whereas when it is occupied its population size follows discrete logistic growth with intrinsic rate r and carrying capacity K . Denoting abundance by $N_i(t)$, post-growth abundance is

$$N_i^{\text{grow}}(t+1) = N_i(t) + r N_i(t) \left(1 - \frac{N_i(t)}{K}\right).$$

Management then removes a fraction h_a of the population, so that

$$N_i(t+1) = N_i^{\text{grow}}(t+1) (1 - h_a).$$

This formulation captures the idea that control reduces numbers but does not eliminate underlying demographic regulation.

Occupancy is updated stochastically based on local abundance and spatial context (Figure ??).

When a site is occupied, extinction probability decreases with abundance according to

$$p_{\text{ext}}(N_i) = \exp(-\alpha N_i),$$

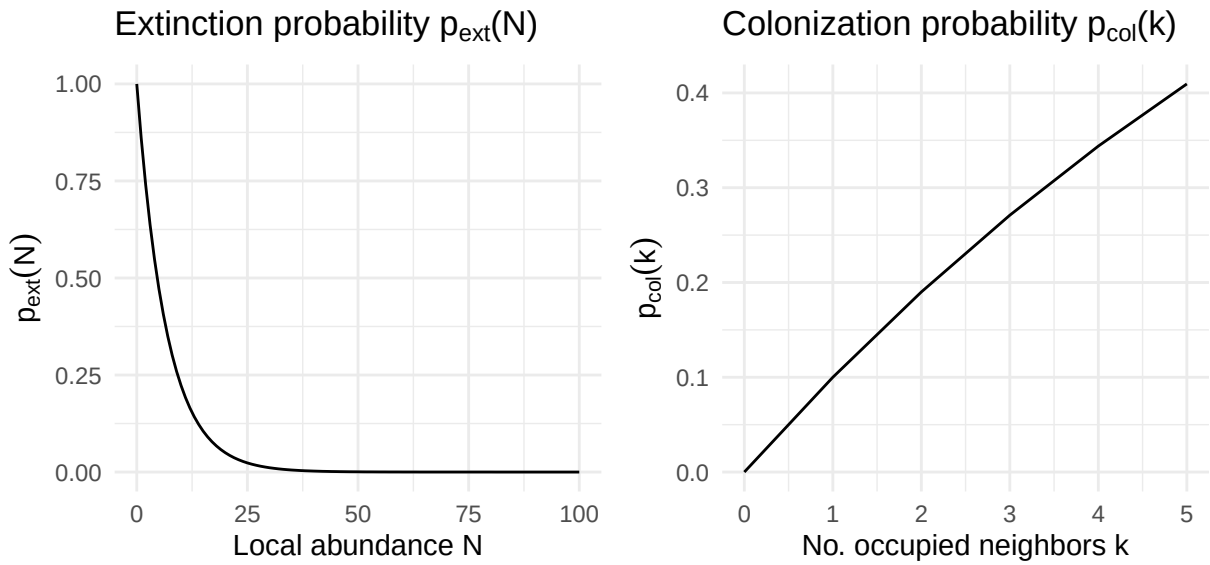
so that small populations are more vulnerable to disappearance. When a site is empty, colonization probability increases with the number of occupied neighbors

$$k_i(t) = \sum_{j \neq i} A_{ij} X_j(t),$$

228 through

$$p_{\text{col}}(k_i) = 1 - (1 - \beta)^{k_i}.$$

229 Together, these rules link local demography with spatial dynamics and create feedbacks between
230 sites.



231
232 Because the transition probabilities induced by these ecological rules are analytically intractable,
233 we approximate them using repeated simulations. Starting from a given state and management
234 action, we simulate the ecological dynamics many times and record the frequency of each
235 possible next state. In practice, the estimated probability of transitioning from state s to state s'
236 under action a is the proportion of simulations that lead to s' :

237 This Monte Carlo approach allows the decision model to incorporate nonlinear and stochastic
238 ecological processes without requiring an explicit analytical expression for transition probabili-
239 ties.

240 Management decisions are evaluated through a cost function that balances ecological damages

241 and intervention effort (Figure 5). At each time step, total cost is defined as

$$C(s, a) = c_d \sum_{i=1}^n N_i(t) + c_a(a),$$

242 where c_d represents the damage cost per individual (here $c_d = 1$) and $c_a(a)$ is the management
243 cost associated with action a . In this example, we consider three possible actions corresponding
244 to no control, moderate control, and strong control, with costs

$$c_a(a) \in \{0, 5, 10\}.$$

245 The reward used in the dynamic programming recursion is defined as the negative of this cost,

$$R(s, a) = -C(s, a),$$

246 so that maximizing expected reward is equivalent to minimizing long-term total costs.

247 The resulting policy is illustrated in Figure 6. When few sites are occupied and populations
248 remain small, limited intervention is often sufficient because damages are modest and extinction
249 may occur naturally. As more sites become occupied or local abundances increase, stronger
250 control becomes optimal in order to reduce future spread and prevent escalating impacts.
251 Temporal variation in recommended actions arises because the finite planning horizon reduces
252 the benefit of aggressive intervention late in the time period.

253 Comparing forward simulations under the optimal policy with simulations under constant
254 strategies highlights the value of adaptive management (Figure 7). The optimal strategy typi-
255 cally reduces cumulative costs by applying effort strategically in time rather than maintaining a
256 fixed level of control.

257 Spatial trajectories further illustrate how coordinated management alters invasion dynamics

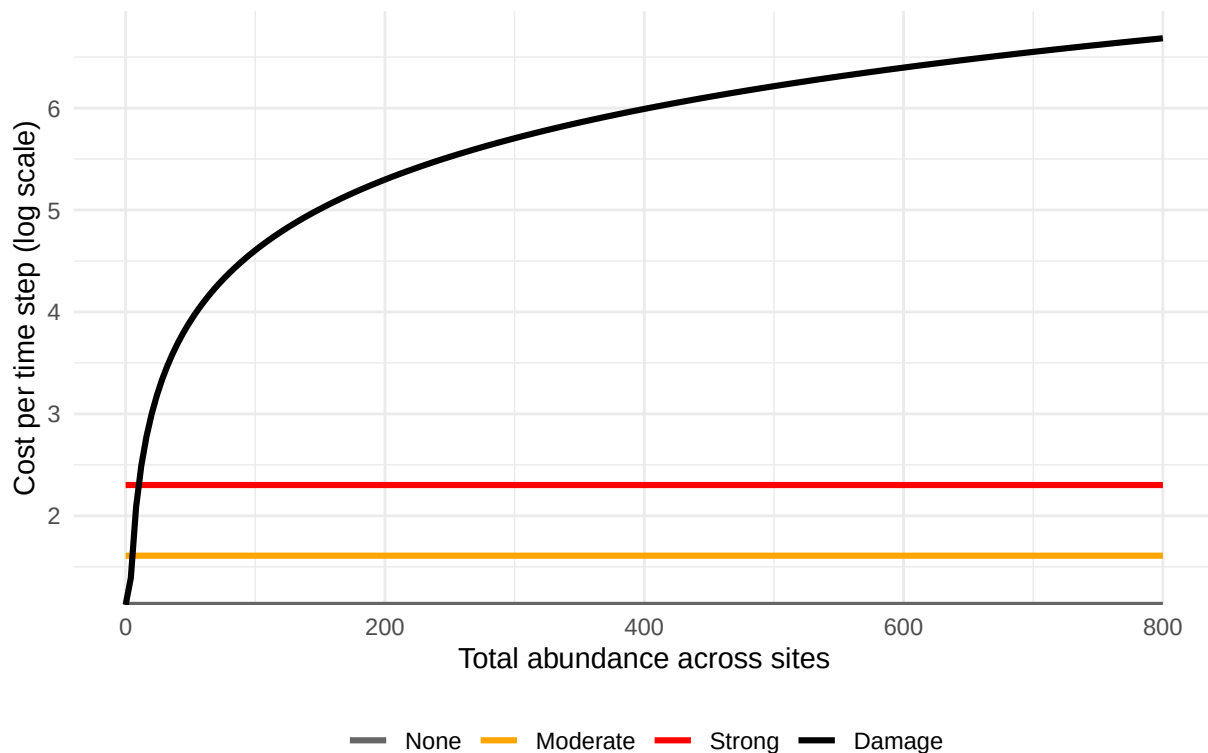


Figure 5: Representation of the cost components on the log scale. Damage costs increase linearly with total abundance across sites, whereas control costs are fixed action-specific intercepts (none/moderate/strong). Stronger control increases immediate spending but can reduce future abundance and damages.

across the network, with reductions in abundance and occupancy emerging progressively across sites (Figure 8).

In perspective, say that this can be used to model competition or predator-prey, or host-pathogen / disease thing. See SDP-species-interactions/ in Olivier biblio repo.

Accounting for observation error w/ POMDP (Cassie)

Illustrate POMDP.

<https://besjournals.onlinelibrary.wiley.com/doi/10.1111/2041-210X.14374> <https://besjournals.onlinelibrary.wiley.com/doi/10.1111/2041-210X.13692>

In an ideal world, invasive species management would be based on perfect knowledge of the state of the system. Yet ecological systems are inherently difficult to monitor, leading to imperfect knowledge of the system and its state. While ongoing monitoring can reduce uncertainties

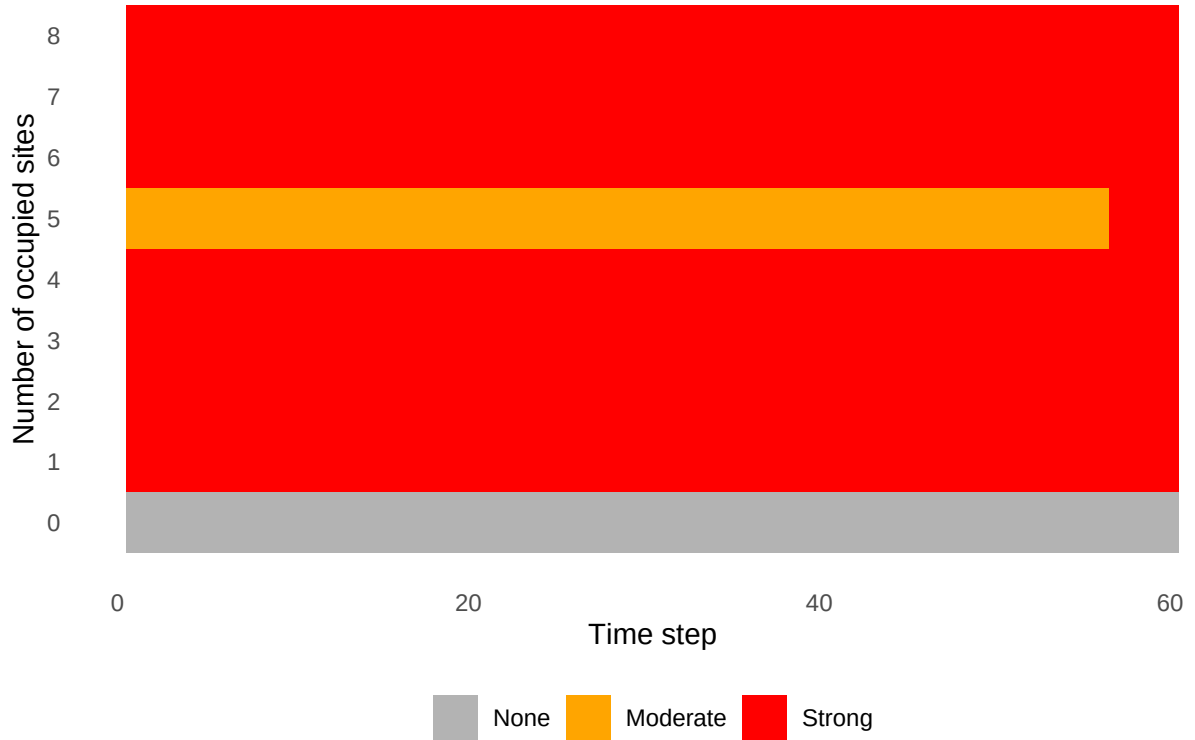


Figure 6: Optimal global action as a function of time and invasion level (measured as the number of occupied sites). Colors indicate the action recommended by the finite-horizon SDP (None/Moderate/Strong). This visualization highlights how stronger control becomes optimal as the invasion occupies more sites, and how recommendations may soften near the end of the planning horizon.

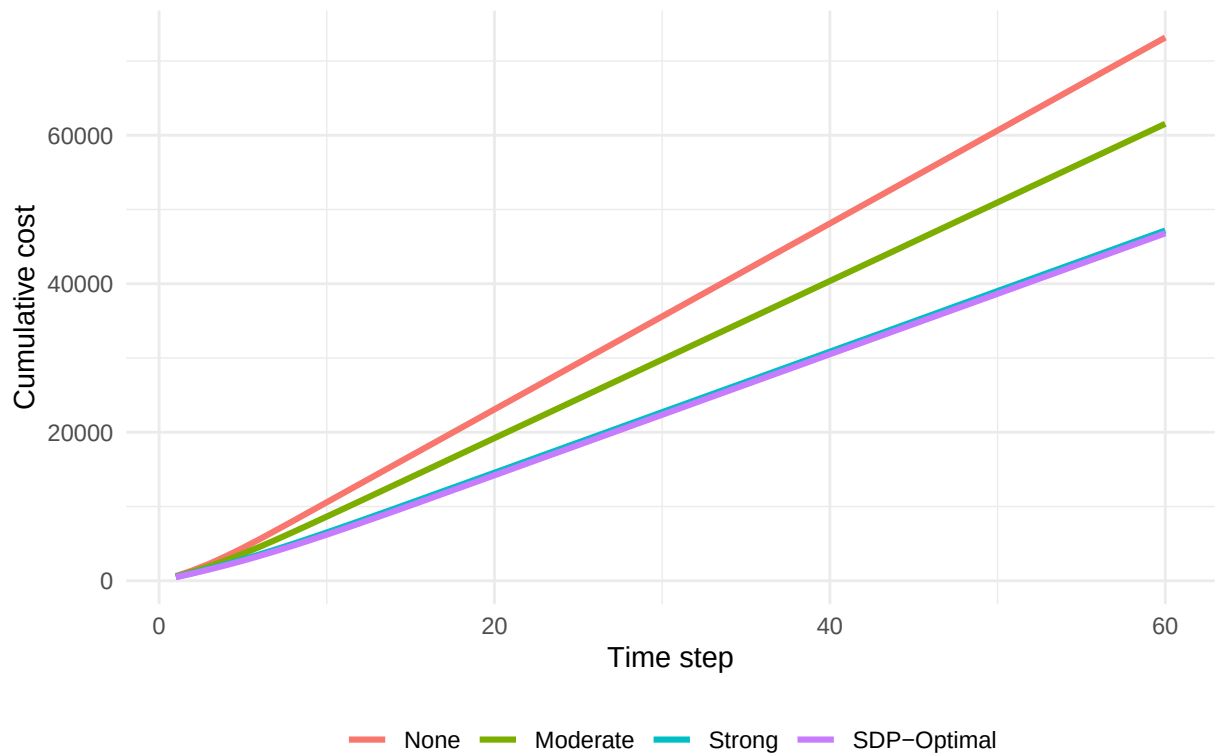


Figure 7: Cumulative costs over time under constant strategies (none, moderate, strong) versus the SDP-optimal policy. Costs combine damages proportional to total abundance and an action-specific control cost.

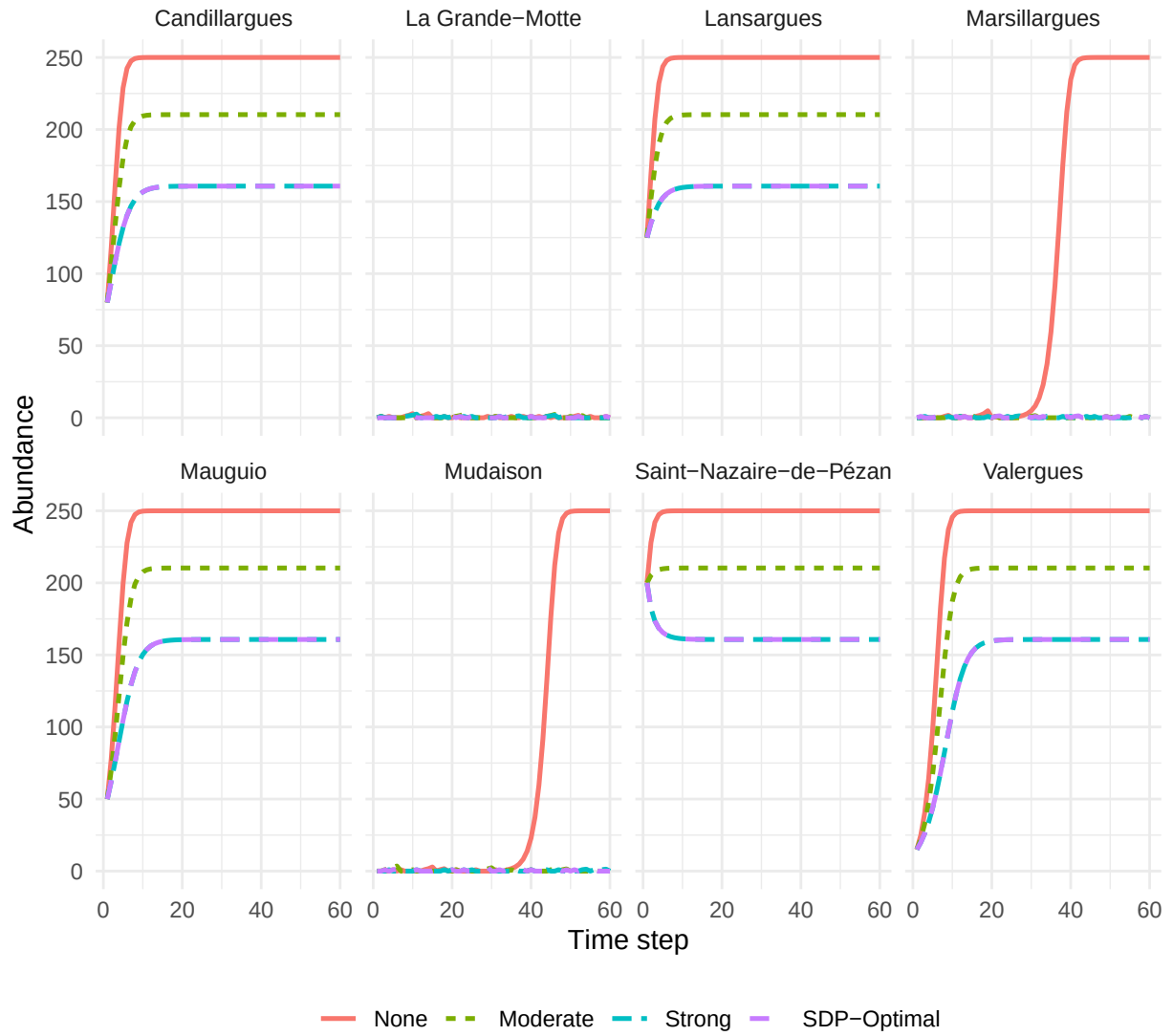


Figure 8: Site-level abundance dynamics under constant strategies and the SDP-optimal policy. Each panel corresponds to a site; trajectories highlight how spatial heterogeneity and network-driven recolonization can create uneven benefits of control across locations.

associated with the state of the system, invasive species management often requires timely decisions. In these cases, partially observable Markov decision process (POMDP) models can be used. POMDPs build on MDPs, dealing with uncertainty in both the model and state of the system. However, POMDPs are more computationally and mathematically challenging to solve due to the continuous and high-dimensional state space. A detailed primer on developing POMDPs for ecological problems already exists (Chades et al 2021), so here we will keep our description simple.

POMDPs differ from MDPs by including an observation state (O), observation function (Z), and initial belief state (b_0). The observation state includes the set of observations perceived by the manager, while the observation function represents the conditional probability of a manager observing o' given that action a lead to state s' . In cases where state uncertainty cannot be reduced, the observation function is independent from the actions taken, and thus dependent only on state s' . The initial belief represents the believed state of the system at a given time.

Because the states are themselves unobservable, the state of the system must be tracked with belief states. At any point in time, the actual (unobservable) state of the system influences the immediate returns, state transitions, and observations, but do not influence actions. The belief state is updated based on observations, which subsequently inform the action(s) to be selected. Lastly, actions drive the state transitions, rewards, and may influence the observations.

[I think having a diagram to show the difference between MDP and POMDP, like Figure 1 in Chades et al 2021 or Fig 2-3 in Williams & Brown 2022, could be helpful] [Also from Chades et al 2021: “In the language of statistics, POMDPs are models that describe optimal control of hidden Markov models (HMMs). In this framing, the partially observed states of the model are latent variables that are inferred from the (fully observable) observation states.”]

Chades et al (2021) identified three types of management problems that POMDPs can be used to solve: - Exploiting current knowledge or reducing uncertainty: Classic dilemma where

managers have to choose between investing in reducing state uncertainty or exploiting current knowledge. - Managing under imperfect detection: Assumes that managers have no means of reducing uncertainty, but managers can invest in management actions for which effectiveness varies across the state space. - Adaptive management: Where the dynamics of the system are not observable. The authors included an example for the first type of problem from a conservation perspective. Here, we'll focus on an example for managing under imperfect detection, continuing with our coyote case study.

Going back to a simple example in invasive species management, imagine we have Occupancy model with uncertain detectability

The invasive species can either be established or absent, with possible observations defined as present or absent denoting if the species was observed in a given time step.

Reducing model uncertainty w/ adaptive management (Abby)

Decision theoretic methods can be used to make control decisions not only in the face of uncertainty, but also to reduce uncertainty over time. Adaptive management, or “learning by doing”, involves making decisions to achieve a management objective while simultaneously learning about system dynamics to improve future management success ([Holling 1978](#), [Walters 1986](#), [Conroy and Peterson 2013](#)). Adaptive management approaches differ based on whether they learn only from the past (passive adaptive management) or anticipate future learning (active adaptive management) ([Chadès et al. 2017](#)).

Here we apply adaptive management strategies to inform control decisions for a single coyote population and to learn about and reduce uncertainty in the per-capita growth rate. We will use the following discrete-time population model, where r is the intrinsic rate of growth; K is the carrying capacity; $q(u)$ is the harvest rate as a function of action u ; R is the per-capita growth rate; m describes whether the per-capita growth rate is linear, concave, or convex; and ϵ

318 is log-normally distributed noise:

$$N_{t+1} = N_t(1 + R(N_t)) - q(u) \times N + \epsilon$$

319

$$R(N_t) = r \times (1 - (N_t/K)^m)$$

320 If $m > 1$, most density—dependent change occurs at high population levels (close to the
321 carrying capacity), which is theoretically common for species with life history strategies typical
322 of large mammals. If $m < 1$, most density-dependent change occurs at low population levels,
323 which is theoretically common for species with life history strategies typical of insects and some
324 fishes ([Fowler 1981](#)).

325 We will consider a model set with five possible values of m . This model bounds, but does not
326 contain, the true model we will use to simulate dynamics in the passive adaptive management
327 approach (Figure 1A). These different values of m result in different optimal policies when
328 system dynamics are known perfectly (Figure 1B).

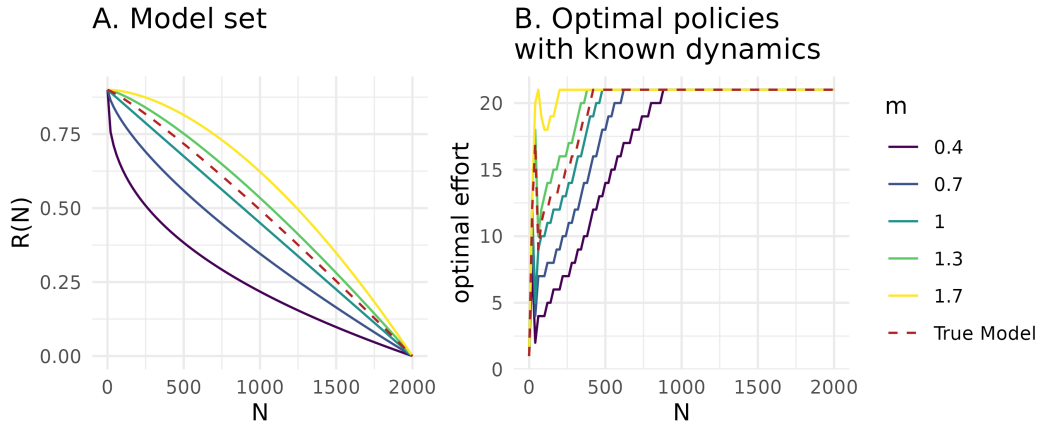


Figure 9: A. Model set representing structural uncertainty in the per-capita growth rate. B. Optimal state-dependent removal actions assuming perfectly known population dynamics, solved with stochastic dynamic programming. In both panels, colors correspond to different values of m in the model set, and the red dashed line corresponds to the true model dynamics used for simulation in the passive adaptive management approach.

Value of Information

For a decision maker, uncertainty may not be important to resolve; learning may not be inherently valuable if it does not result in improved management outcomes. A Value of Information (VoI) analysis *a priori* quantifies the importance of reducing uncertainty, or the expected improvement on management objectives if uncertainty were resolved (Raiffa and Schlaifer 2000). If the VOI is high, an adaptive management approach is typically justified.

The expected value of perfect information (EVPI) is measured as the difference between 1) PI , or the expected value once uncertainty has been resolved because the optimal action is chosen after knowing which model best describes the system, and 2) NL , or the expected value in the face of uncertainty because the optimal action is chosen with no learning about the system (Runge et al. 2011). The difference between these terms, $EVPI = PI - NL$, is the expected improvement in management performance due to acquiring perfect information.

341 To calculate EVPI for a sequential decision problem, we first use SDP to calculate the optimal
 342 state-dependent policy with assumed system dynamics, $\pi^*(s)_k$, for each model k in the model
 343 set K , given each transition matrix, P_k . We also calculate the bet-hedging strategy, or the optimal
 344 policy given uncertain dynamics, $\pi^*(s)_{NL}$. This bet-hedging strategy is calculated with the
 345 transition matrix P_{NL} , representing the average of the transitions of each model, P_k , weighted
 346 by each model belief, b : $P_{NL}(s_{t+1}|s_t, a_t) = \sum_{k \in K} b(k)P_k(s_{t+1}|s_t, a_t)$.

347 Using the policies calculated with known dynamics, $\pi^*(s)_k$, and unknown dynamics, $\pi^*(s)_{NL}$,
 348 we find the EVPI by first simulating i replicated population trajectories assuming each model
 349 k , applying both policies with known and unknown dynamics, and calculating the value
 350 at time t over time horizon T . The value associated with applying the policy with known
 351 dynamics is $V_{k,t,i}(s_t, a_t|a_t = \pi^*(s)_k)$, and the value associated with applying unknown dy-
 352 namics is $V_{k,t,i}(s_t, a_t|a_t = \pi^*(s)_{NL})$. The EVPI is therefore $E_{s,k,t,i}[V(s_t, a_t|a_t = \pi^*(s)_k)] -$
 353 $E_{s,k,t,i}[V(s_t, a_t|a_t = \pi^*(s)_{NL})]$.

354 In the context of coypu regulation, the EVPI represents the expected improvement on the
 355 management objective if uncertainty about the per-capita growth rate was reduced. We calculate
 356 the EVPI for a model set, K , where $m \in \{0.4, 0.7, 1, 1.3, 1.7\}$ for $I = 200$ iterations over a $T = 40$
 357 time horizon. We assume equal belief in each model. The expected value once uncertainty has
 358 been resolved, PI , is -1730.4, the expected value with no learning, NL , is -1787.8, and the EVPI is
 359 $PI - NL = 57.4$. It is therefore valuable to learn about the population dynamics and per-capita
 360 growth rate, as this learning is expected to improve performance on the management objective.

361 Value of information analyses are typically applied for static, one-time conservation decisions,
 362 rather than sequential problems, often in the context of understanding when to act and when to
 363 delay action until after further research (Bolam et al. 2019). To better understand the value of
 364 information in the context of adaptive management, future work should investigate quantifying
 365 VOI by iteratively resolving uncertainty, requiring evaluating the gain of implementing an

366 adaptive policy rather than a single action (Chadès et al. 2017).

367 **Passive adaptive management**

368 Passive adaptive management involves learning from the outcomes of previous management
369 actions to update knowledge about system dynamics and improve management performance
370 over time. Decisions at each time step are optimized assuming that current knowledge of the
371 system will not change into the future. Learning therefore occurs by monitoring the effect of an
372 implemented action, rather than during the optimization procedure (Chadès et al. 2017).

373 Solving a passive adaptive management problem includes an implementation step and an
374 updating step at every time point t , based on the current belief in model k , $b_t(k)$. The MDP must
375 be solved at every time step because the transition matrices change as the belief changes. During
376 the implementation step, the optimal action at time t is selected using a weighted averaging
377 approach, where the transition probabilities are averaged across all models with the weights
378 given by the current belief, $b_t(k)$ (Williams et al. 2011). This can be solved using stochastic
379 dynamic programming:

$$V^*(s_t|b_t) = \max_{a \in A} \left\{ \sum_{k \in K} b_t(k) r(s_t, a_t) + \gamma \sum_{s_{t+1}} \left\{ \sum_{k \in K} b_t(k) P_k(s_{t+1}|s_t, a_t) \times V^*(s_{t+1}|b_t) \right\} \right\}$$

380 During the updating step, the resulting state, s_{t+1} , after taking action a_t is observed, and belief,
381 b_{t+1} , is updated according to Bayes' rule:

$$b_{t+1}(k|s_t, a_t, s_{t+1}, b_t) = \frac{P_k(s_{t+1}|s_t, a_t) b_t(k)}{\sum_{k \in K} P_k(s_{t+1}|s_t, a_t) b_t(k)}$$

382 For coypru control decisions, we solve the passive adaptive management problem by updating
383 the belief about the coypru population dynamics to improve management performance over

time. We begin with an equal belief, $b_{t=1}(k) = 1/5$, across the five models in the model set (Figure 1A). We iterate between 1) calculating the optimal action given the current belief state, 2) implementing the optimal action and simulating the next state, s_{t+1} , given the true population dynamics, and 3) updating the belief state based observing the next state given action a_t . Here we assume the state is perfectly observed. We simulate the application of passive adaptive management over 30 time steps. The belief in each model changes over time (Figure 2A), resulting in the highest belief for the two models closest to the true population dynamics $m = 1$ and $m = 1.3$ (Figure 1A). Through updating the belief state over time and implementing the optimal action given the current belief state, the management performance improves over time (Figure 2B).

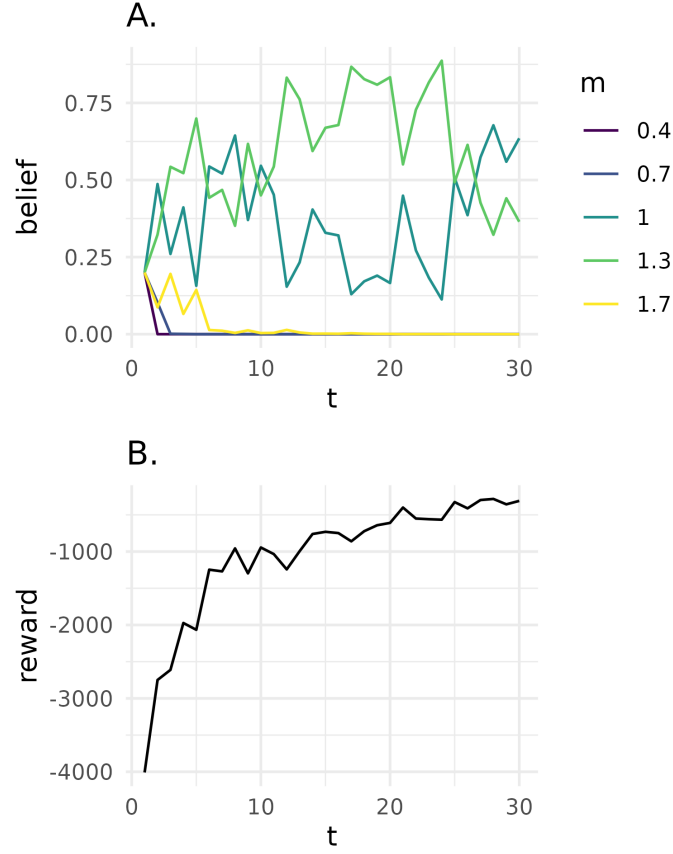


Figure 10: A. Belief state for each model k at time t . Colors correspond to different values of m , representing structural uncertainty in the per-capita growth rate. B. Reward, $r(s_t, a_t)$, at time t in passive adaptive management problem.

Active adaptive management

In contrast to passive adaptive management, active adaptive management does not rely only on past experience and instead explicitly accounts for future learning opportunities to maximize the chance of achieving the objective. Instead of an optimization at every time step as in passive adaptive management, the optimization occurs once and requires that the trajectory of belief b_t to b_{t+1} and state transitions be calculated for all time steps (Chadès et al. 2017). The optimal value function V^* that characterizes the performance of a policy is a function of both the state of the system s_t and belief over the models b_t :

$$V^*(s_t, b_t) = \max_{a \in A} \left\{ \sum_{k \in K} b_t(k) r(s_t, a_t) + \gamma \sum_{s_{t+1}} \left\{ \sum_{k \in K} b_t(k) P_k(s_{t+1} | s_t, a_t) \times V^*(s_{t+1}, b_t) \right\} \right\}$$

Active adaptive management with model uncertainty can be solved with many methods, including those developed for dealing with partial observability (Williams et al. 2011, Chadès et al. 2017). Here, a POMDP is characterized by a tuple $\langle X, A, O, P, Z, r, \gamma \rangle$, where $X = S \times K$ represents the factored state space with S denoting the possible conditions of the system and K denoting the model set. The unknown model is an unobserved state variable that must be inferred through observation. We can solve the POMDP with the Successive Approximations of the Reachable Space under Optimal Policies (SARSOP) algorithm (Kurniawati et al. 2008). With this algorithm, the value function V^* associated with the optimal policy π^* can be approximated by a convex, piecewise-linear function $V(b) = \max_{\alpha \in \Gamma} (\alpha \cdot b)$, where Γ is a finite set of α -vectors and b is the discrete vector representation of a belief (Kurniawati et al. 2008).

For coypru control decisions, we solve the active adaptive management problem by augmenting the state space with the belief in two models where $m \in \{0.7, 1.3\}$. We choose a smaller model set than in the passive adaptive management problem due to the computational cost associated with exploring the belief and state spaces over the entire time period (i.e., “curse of dimensionality”). We assume a uniform belief across models and assume perfect observation of the state (i.e., $o_t = s_t$).

We also use the coypru example to demonstrate the “dual control problem” (Walters and Hilborn 1978). A management action altering a poorly understood system can produce two types of benefits: 1) short-term payoffs and 2) learning about the system so as to improve payoffs in the long-term. There is a trade-off between these two benefits, as large disturbances allow for rapid learning yet may incur high costs; the challenge is to optimally balance these benefits. Here we explore the dual control problem by selecting two different discount factors, $\gamma \in \{0.95, 0.75\}$.

424 The higher of which represents a decision maker who values long-term rewards more highly,
 425 and the lower of which represents a decision maker who values short-term rewards more highly.
 426 The active adaptive management solution describes the optimal action (a_t , removal effort),
 427 given the state (s_t , coyup abundance), initial belief ($b_{t=1}$, uniform across models), and discount
 428 factor (Figure 3). The solution differs depending on how future rewards are discounted in the
 429 optimization. At low coyup abundance, the optimal removal effort increases as the discount
 430 factor increases. With a higher discount factor, future system states are valued more highly; even
 431 though a higher removal effort is costly today, this higher removal effort will accelerate learning
 432 to yield higher rewards in the long term (Figure 3A). In contrast, with a lower discount factor,
 433 active learning becomes less useful, as the price paid for not following the passive-adaptive
 434 (optimal right now) solution exceeds the information gained from learning (Figure 3B).

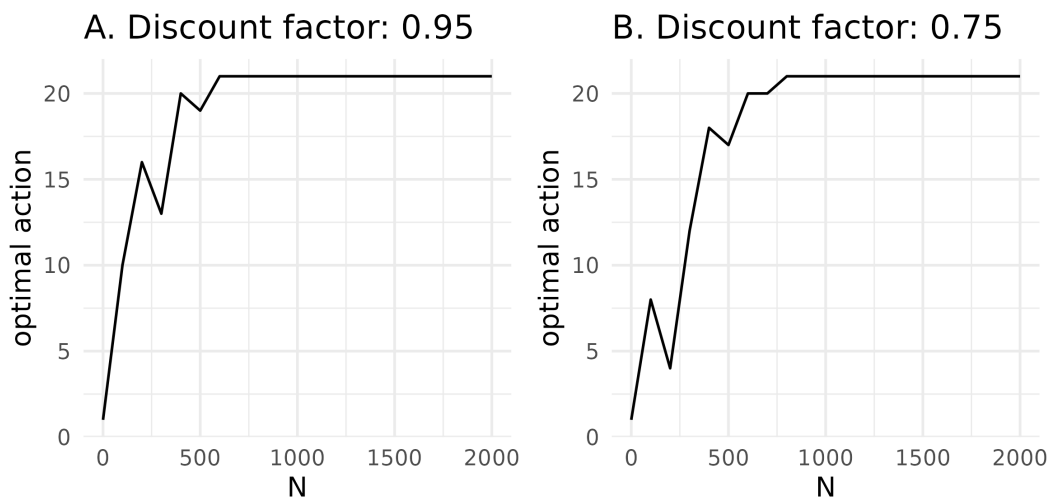


Figure 11: A. State-dependent active adaptive management solution, given an initial uniform belief across models and a higher discount factor representing a decision maker who values long-term payoffs more highly. B. State-dependent active adaptive management solution, given an initial uniform belief across models and a lower discount factor representing a decision maker who values short-term payoffs more highly.

435 On a single species ; check out Conroy's book, there is a nice figure illustrating AM.

436 <https://onlinelibrary.wiley.com/doi/10.1002/ece3.71176>

437 **Discussion (all)**

438 **Main messages**

439 How is SDM/SDP relevant.

440 **Limitations**

441 Curse of dimensionality.

442 **Perspectives**

443 **Acknowledgments**

444 **Data availability statement**

445 Data and code are available at <https://github.com/oliviergimenez/SDPaper>.

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